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Concentrator operational modes in response to geological variation

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ABSTRACT

Geological variation may be managed by alternating between modes of operation. These modes provide an integrated response to changes in feed mineralogy, and other operational conditions within the mineral value chain. The decision to alternate between modes depends on current and forecasted stockpile levels. Moreover, the optimization of stockpile thresholds that would trigger a mode change is related to the classical RQ problem from inventory theory. Particularly for concentrators that are designed for blended feeds, there may be production intervals in which a particular ore class is at risk of stockout. Indeed, geological uncertainty causes stockout risk, which should be balanced with other key performance indicators, including throughput and recovery, and is a context for multiobjective optimization. Computations are presented in which a concentrator that had originally been designed for a certain ore blend, after years of successful operation, now requires alternate modes to realign its operations with geological forecasts.

1. Introduction

Modes of operation are fundamental to the integrated management of mining and extractive metallurgical complexes (Navarra et al., 2017a, 2017b). Operational modes are conceived with respect to system-wide metrics, rather than local metrics that only consider isolated unit operations. Thus, when a multicomponent system operates under a given mode, each unit operation follows a set of instructions that is coordinated on the system-wide level. For instance, the operating parameters of a comminution circuit should be altered in coordination with the incoming feed as well as the downstream separation processes. More widely, the multidisciplinary field of geometallurgy coordinates geological attributes with mining and processing decisions (Lotter 2011; Lotter et al., 2011; Navarra et al., 2018), thereby adapting the concepts of integrated management, including operational modes.

In previous work, Navarra et al. (2017a) extended the strategic mine planning algorithm developed by Lamghari et al. (2014), to evaluate the benefit of having an elaborate concentrator that is able to accommodate different types of ore, rather than a low-cost concentrator that may not be well-suited for all ores; the more elaborate plants can be modelled as having more operational modes. By comparing the net present value (NPV) of the mining project with and without the additional modes, the strategic planning algorithm determines whether the benefit of the additional modes could offset the increased capital expense of the more elaborate plants. Furthermore, Navarra et al. (2017a) observed that concentrator design decisions are coupled with the initial strategic mine plan, as changes in the concentrator design correspond to changes in the optimal mine plan. Moreover, the concentrator design and strategic mine plan are both subject to geological uncertainty; the exact nature and distribution of the ore is uncertain, especially in the early stages of the mine life, as a result of the sparsity of drill core samples.

The strategic planning algorithms of Navarra et al. (2017a, 2018), and similarly those of Ravenscroft (1992), Lamghari et al. (2014), and others (Newman et al., 2010), are limited in the degree of realism that can be afforded to individual unit operations. The operational modes are regarded as black boxes, characterized by time-averaged data, such as mining cost per ton of rock, milling cost per ton of ore, recovery, throughput, etc., without any detail of tactical phenomena such as short-term feed variation, equipment breakdowns or seasonal conditions (Marois et al., 2018). The strategic analysis can only describe time-averaged trends, demonstrating, for instance, that certain modes become favoured as different parts of the deposit are excavated. However, it does not describe the dynamic conditions that would trigger a change in operational mode (Wheeler and McGinty, 2015). In particular, if a new region of an orebody is to be excavated, then it may be beneficial for the concentrator to alternate between different modes, especially if the new region has categorically different geometallurgical

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attributes than the regions that are already being exploited; however, this depends on the ore body knowledge that is available at the time of decision-making, including the degree of uncertainty. Circuit designs tend to prioritize the ore characteristics of the early feed, with the possibility of retrofitting later within the mine life, thus to support new operational modes as different ores come into operation. The current paper develops the quantitative framework to optimize the policies that determine when to alternate between modes of operation, which assists in concentrator operation, as well as concentrator design and re-engineering projects. The approach is an adaptation of concepts used in multiobjective optimization and inventory theory (Miettinen, 1998; Chavas, 2004; Winstin and Goldberg, 2004; Chuk et al., 2018), as described in the following section.

There are numerous studies showing that changes in ore hardness and mineralogical characteristics affect throughput and liberation potential, which should be responded with changes in circuit operating parameters; these studies have been reviewed by Mwanga et al. (2015), which cites the empirical work at the Collahuasi Mine, in northern Chile (Alruiz et al., 2009). The advancements at Collahuasi incorporated a significant number of Bond Index and JK drop tests into the analysis of their geological model, which became the basis for integrated mine-to-mill management. This work was extended through an industry-academic collaboration to incorporate flotation models (Suazo et al., 2010), which influenced the later work of Navarra et al. (2017a, 2018) in strategic mine planning. However, the potential to float sulfide minerals depends on the surface oxidation of the sulfides (Bicak and Ekmekci, 2012), and gangue characteristics (Tungpalan et al., 2015), that can negate the benefit of increased liberation. Grinding circuit optimization can nonetheless produce a desired particle distribution (Chimwani et al., 2013) which, given appropriate ore sorting and pre-concentration (Lessard et al, 2016; Carrasco et al., 2017; Hesse et al., 2017), can decrease energy costs without diminishing recovery.

Comminution continues to receive attention from the research community because it is a dominant operating cost (Sayadi et al., 2014; Lotter et al., 2018). The industrial survey of Curry et al. (2014) determined that comminution comprises 40% of operating costs, in which grinding is the major component. In practice, however, the greatest potential for optimization is realized when communition is considered within a system-wide context, that considers the comminution and the downstream separation, in conjunction with mineralogical characterization. In particular, Lotter et al. (2018) describe the greenfield development of the Montcalm Ni-Cu project in northern Ontario in which the flowsheet was modified to include a coarse primary grind followed by a regrind, which favours a sequential concentration of pentlandite before chalcopyrite; the resulting process can then balance the grinding intensity between the primary and the regrind, in function of the incoming pentlandite versus chalcopyrite. Lotter et al. (2018) also present a number of brownfield projects, including the retrofit of the Amandelbut plant in South Africa that concentrates PGM. This plant had suffered from ultrafine PGM losses into the silicate tailings, in combination with incomplete PGM liberation from the silicate grains. The plant was then retrofit to regrind both the rougher tailings and the cleaner feed, while the primary circuit liberates the silicates from a chromite matrix. In addition to the cases described by Lotter et al. (2011, 2018), and the work at Collahuasi (Alruiz et al., 2009; Suazo et al., 2010), Lishchuk et al. (2015) present two more cases studies: firstly, a geometallurgical model was used to simultaneously improve the mine planning and downstream processing of the Mikheevskoye Cu-porphyry deposit in Russia; secondly, a framework was developed to optimize the plant performance at the Malmberget iron ore deposit, in Sweden. Each of these cases demonstrate the importance of integrated management within the mineral value chain, and are part of the motivation for the current paper.

The aforementioned case studies consider time-averaged steady flows. However, it is the short-term risk of dynamic surging, rather than time-averaged flows, that determines equipment capacity and operational policies (Chavas, 2004; Wheeler and McGinty, 2015). Incidentally, some of the most recent work in mine-to-mill modelling considers the dynamic behaviour of equipment and systems (Srivastava et al., 2018; Nageshwaraniyer et al., 2018). The model of Srivastava et al. (2018) focuses on the localized control of semi-autogenous grinding (SAG), but could be integrated into a system-wide framework similar to that of Nageshwaraniyer et al. (2018) to develop broader-reaching operational modes. Indeed, Nageshwaraniyer et al. (2018) have created a discrete event simulation (DES) framework to quantify the benefit of a spectral imaging-based tracking of ore types; this requires a representation of all affected components, to compare operations with and without the tracking system. More generally, the justification of novel technologies requires some representation of how the enhanced system will be operated; this is exemplified by Pamparana et al. (2017) that quantify the benefit of coupling a SAG mill with solar energy generation. Considering the competitive nature of the mining industry, the sustainability of mines will depend on the systemic coordination of automated technology (Batterham, 2017), which we formulate as alternating operational modes (Navarra et al., 2017a, 2017b). Sustainability analysis requires the balancing of opportunities and risks, over different timescales, which is closely related to multiobjective optimization (Miettinen, 1998). Interestingly, Chuk et al. (2018) present mine-to-mill analysis through a multiobjective framework that balances the NPV over the life of a Au-Ag mine, while maintaining stable Au production. Chuk et al. (2018) describe fundamental concepts such as Pareto optimality in the context of blending strategies, but do not relate dynamic stockpile management to alternating operational modes.

The optimal functioning of a concentrator depends on stable feed streams. Indeed, ore stockpiles are a buffer against short-term geological variation, as blending can be used to stabilize the concentrator feed. Mode changes are triggered by observed and forecasted changes in stockpile levels, thus providing an integrated (system-wide) response to geological variation. These tactical triggers are closely connected to a classical problem in inventory theory: the so-called RQ problem (Winstin and Goldberg, 2004). Through this connection, formulations that are well-established in inventory theory could be adapted to ore stockpile management, benefitting from advances in DES frameworks, and simulation-based optimization (Kleijnen and Wan, 2007).

2. Pareto optimality and stockout risk

Engineering systems are subject to a set of key performance indicators (KPIs) that simultaneously measure their compliance with several objectives. The most basic KPI is usually a tally of the operational costs per unit of production, which are to be minimized. Other KPIs can include productive metrics such as throughput, or risk-aversion metrics such as customer satisfaction (i.e. satisfied customers are less likely to find a new supplier). In practice, an increase in operational costs can only be justified if it coincides with improvements in other indicators. However, the different KPIs are not necessarily comparable, in a direct quantitative sense, which motivates the concept of Pareto optimality, also known as Pareto efficiency (Miettinen, 1998; Chavas, 2004; Chuk et al., 2018).

A system is said to be Pareto optimal with respect to a set of objectives, if any conceivable improvement with respect to one of the objectives would be at the expense of at least one other objective. This notion is fundamental to multiobjective optimization, and is especially relevant in the early phases of an engineering project (McMeekin et al., 2015). Even when various critical risks have been identified in an early phase, their impact on the profitability and sustainability of the system may not be fully understood without a detailed analysis of the later phases. Meanwhile, certain options may be immediately disfavoured, if their increased costs and risks are not counterbalanced by the possibility of improvements in other metrics. In particular, if there are two configurations, I and II, such that there are no metrics by which I is

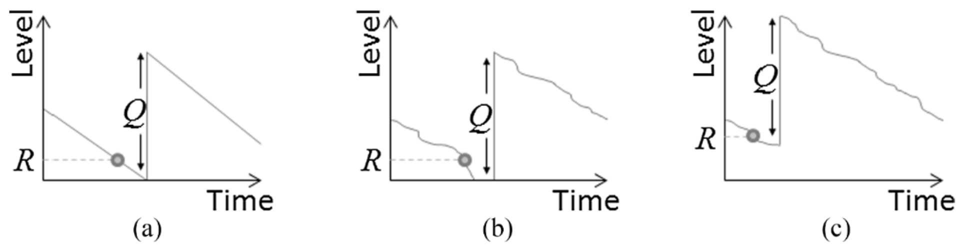


Fig. 1. Adjustment of R to mitigate stockout risk. (a) Deterministically optimum R value applied to a deterministic RQ problem, (b) Deterministically optimum R value applied to a stochastic RQ problem leads to stockout, (c) Increased R value mitigates stockout risk but increases inventory handling and storage costs.

Fig. 2. Pareto frontier describing the tradeoff between total operational cost per unit and demand that is immediately satisfied.

significantly better than II, and at least one metric by which II is significantly better than I, then configuration II dominates configuration I; therefore, configuration I can be immediately discarded in favour of configuration II. In the mineral context, two mine plans may be expected to produce similar NPV, but one of these plans may be more robust in response to geological variations (Chuk et al., 2018); the less robust plan is discarded, because it is dominated by the more robust plan.

The RQ problem is a classic problem from inventory theory (Winstin and Goldberg, 2004), and is fundamental to both risk management and multiobjective optimization. As an inventory level for a certain consumable (or saleable) item is being diminished, it eventually crosses the reorder point R (Fig. 1), at which time an order is sent out, so that a resupply quantity Q may arrive before the inventory falls to zero, i.e. before there is a stockout event. The duration of time, between when the order is initially sent and when the resupply is finally received, is known as the leadtime. In a simplified analysis, the leadtime and consumption rate can both be regarded as deterministic (Fig. 1a); a deterministic analysis suggests that

$$R = rl \tag{1}$$

in which l is the expected leadtime, and r is the expected consumption rate. Higher inventory levels tend to involve higher operating expenses to manipulate and secure the items, and potentially higher capital costs for handling equipment and storage space. Therefore, Eq. (1) postpones the reordering until the last possible moment, to avoid excess inventory.

In practice however, the consumption rate, and possibly the leadtime, may be subject to random variation. Following a reorder event, the consumption may be greater than expected or the leadtime may be longer than expected, hence a risk of stockout (Fig. 1b). In order to mitigate this risk, the reorder point must be raised higher than the deterministic result (Fig. 1c), even though this will increase short-term operational costs; this increase is known as the security stocklevel (Winstin and Goldberg, 2004). Depending on the exact context, the impact of a stockout may not be directly assessable on a common basis with inventory costs, hence a context for multiobjective optimization. For example, in a university lab, it would be unwise to apply Eq. (1) to decide when to order new safety goggles; in this case, if there are frequent stockouts, then certain students may be pressured to run an experiment without adequate eye protection, and there may eventually be a tragic accident. Certainly, the perceived cost of cabinet space is not directly summable with the issue of student safety. In this case, the security stock should be high enough so that students are "never" pressured to forego the use of goggles.

The RQ problem usually implies some flexibility so that the two tactical parameters, R and Q, must be determined simultaneously to minimize operational costs while maintaining acceptable levels in other metrics, such as likelihood of stockouts. The impact of Q on operational expenses is described by Winstin and Goldberg (2004), and is often context-specific, including fixed and variable costs. Nonetheless, the reorder and replenishing events that characterize the RQ problem can be analyzed using a DES software, such as Rockwell Arena©; the resulting framework can then simulate hundreds of days of operation, thereby comparing the operational performance with respect to various KPIs, using different values of R and Q. As demonstrated in the following section, a DES framework can be run in conjunction with an optimization engine, to locate Pareto optimal configurations. The set of Pareto optimal configurations is known as the Pareto frontier (Miettinen, 1998; Chavas, 2004), which determines the (optimal) tradeoffs between different KPIs. In particular, Fig. 2 illustrates the tradeoff between operational costs and the portion of demand that is immediately satisfied (i.e. without suffering from a stockout); this type of curve gives the short-term costs of increased customer satisfaction, even when the medium- to long-term benefits are not fully known. In general, there may be a class of niche customers that are willing to pay more to have this level of assurance, hence justifying the increased operational costs; Pareto optimality is thus related to market segmentation (Boom et al., 2015).

Moreover, a DES framework can be configured to consider more elaborate production problems, including context-specific operational parameters, events and KPIs. Fig. 3 illustrates the alternation between two modes: a consumption mode and a replenishment mode (Winstin

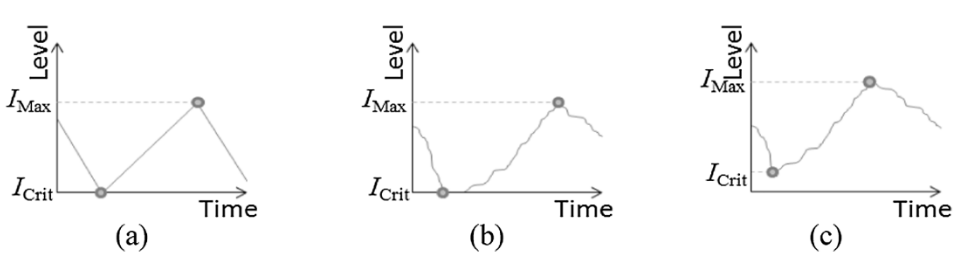


Fig. 3. Adjustment of I_{Crit} to mitigate stockout risk, while maintaining constant (I_{Max}-I_{Crit}). (a) Deterministically optimum I_{Crit} value applied to a deterministic two-mode problem, (b) Deterministically optimum I_{Crit} value applied to a stochastic two-mode problem leads to stockout, (c) Increased I_{Crit} value mitigates stockout risk but increases inventory handling and storage costs.

and Goldberg, 2004). The consumption mode is presumably more profitable than the replenishment mode, with regard to short-term production metrics; however, the consumption mode is not sustainable without the periodic application of the replenishment mode. In this case, there may be two tactical parameters, I_{Crit} and I_{Max}, which are loosely analogous to R and Q, respectively. The I_{Crit} value is the critical inventory threshold that triggers the replenishment mode. Fig. 3 considers that the replenishment mode is immediately activated, however there may be technical reasons why the change would have to be delayed; in the next section, for instance, the mode changes can only be executed during the next plant shutdown. In a purely deterministic context, I_{Crit} may be set to zero (Fig. 3a), however any variation in consumption rate is likely to cause stockouts (Fig. 3b), unless I_{Crit} is raised so as to provide an appropriate security stock (Fig. 3c). Note that in the stochastic setting, I_{Crit} does not define a true minimum because the stocklevel may continue to diminish, even after the replenishment mode has been triggered.

There is a notable comparison between the variation of a consumer retail context (be it an RQ scenario or a two-mode scenario, Figs. 1 and 3, respectively), and that of an ore stockpile. In general, the change in inventory level is given by the incoming rate of supply, minus the outgoing rate of demand. Within the retail context, however, there is usually uncertainty in the rate of demand, as consumer behaviour is subject to statistical variations; in this case, the supplying is comparatively well-controlled. As for the mining context, ore stockpiles are generally subject to geological variation that affects the supplying of ore, while the outgoing concentrator feed is comparatively well-controlled through blending practices. Indeed, the geological variation has an analogous effect to consumer variation, except that the former acts on the supply side, and the latter acts on the demand side. In either case, the net effect is similar: variation in stocklevels leads to stockout risk.

The RQ and two-mode models have wide-ranging applications within mining, manufacturing, retail and, more generally, in risk management. Within these models, the context-specific components must include a description of the recourse actions that are performed in case of stockouts, and how these actions affect the KPIs. The following section considers an adaptation of the two-mode model, which is to manage geological variation by alternating the configuration of concentrator operational modes. The adaptation considers that there are two ore types, Ore 1 and Ore 2, which are blended and processed under two modes, Mode A and Mode B, with feed rates r_A and r_B, respectively. The blended feed for Mode A has predetermined weight-fractions of ores 1 and 2, denoted w_{1A} and w_{2A}, respectively. Likewise, the feed for Mode B is a predetermined blend, described by w_{1B} and w_{2B}. The deposit is expected to provide weight fractions of Ore 1 and Ore 2 that are given by w_{1D} and w_{2D}, respectively. Without loss of generality, Mode A may be more productive than Mode B; however, the perpetual application of Mode A will cause chronic stockouts as $w_{2A} > w_{2D}$. Thus, a sustainable operation requires periodic alternations between the two modes, so that the Ore 2 stockpile may be periodically replenished. In case stockouts do occur, the concentrator must apply contingency modes which are less productive than the regular modes, i.e. Mode A-Contingency is less productive than the regular Mode A, and Mode B-Contingency is less productive than the regular Mode B. r_{ACont} denotes the blended feed rate for Mode A-Contingency, and w_{1ACont} and w_{2ACont} are the weight fractions of Ore 1 and Ore 2 that are fed into Mode A-Contingency, respectively. A similar notation is applied for B-Contingency, as described in Table 1.

By applying a deterministic analysis to balance the concentrator operations (r_A, w_{1A}, w_{2A}, r_B, w_{1B}, w_{2B}) with the geological expectation (w_{1D}, w_{2D}), the time devoted to each mode satisfies the following relationship,

$$\left(\frac{t_A}{t_B}\right) = \left(\frac{w_{2B}w_{1D} - w_{1B}w_{2D}}{-w_{2A}w_{1D} + w_{1A}w_{2D}}\right)\left(\frac{r_B}{r_A}\right) \tag{2}$$

Table 1
Algebraic notation for the adaptation of the two-mode model.

	Mode A		Mode B		Deposit
	Regular	Contingency	Regular	Contingency	
Throughput (t/h)	r_A	r_{ACont}	r_B	r_{BCont}	–
Ore 1 in feed (%)	w_{1A}	w_{1ACont}	w_{1B}	w_{1BCont}	w_{1D}
Ore 2 in feed (%)	w_{2A}	w_{2ACont}	w_{2B}	w_{2BCont}	w_{2D}

in which t_A is the duration of time devoted to Mode A, and t_B is the duration of time devoted to Mode B. The average throughput is then given by

$$r = \left(\frac{w_{1A}w_{2B} - w_{2A}w_{1B}}{\left(w_{2B}\left(\frac{r_B}{r_A}\right) - w_{2A}\right)w_{1D} - \left(w_{1B}\left(\frac{r_B}{r_A}\right) - w_{1A}\right)w_{2D}}\right)r_B \tag{3}$$

Eqs. (2) and (3) can be adapted to describe an operational policy which would perpetually apply the configuration of Mode A, and thus suffer from chronic shortages of Ore 2, and therefore be forced to alternate between the regular Mode A and Mode A-Contingency:

$$\left(\frac{t_A}{t_{ACont}}\right) = \left(\frac{w_{2A}w_{1D} - w_{1A}w_{2D}}{-w_{2ACont}w_{1D} + w_{1ACont}w_{2D}}\right)\left(\frac{r_A}{r_{ACont}}\right) \tag{4}$$

$$r = \left(\frac{w_{1A}w_{2ACont} - w_{2A}w_{1ACont}}{\left(w_{2ACont}\left(\frac{r_{ACont}}{r_A}\right) - w_{2A}\right)w_{1D} - \left(w_{1ACont}\left(\frac{r_{ACont}}{r_A}\right) - w_{1A}\right)w_{2D}}\right)r_{ACont} \tag{5}$$

Eqs. (4) and (5) are identical to Eqs. (2) and (3), except that Mode A-Contingency takes the place of Mode B. The result of Eq. (5) should generally be larger than that of Eq. (3), otherwise Mode B is not viable. In a simple deterministic context, the objective is indeed to maximize throughput. The following section describes a stochastic bi-objective context, to maintain maximal production with minimal stockpile levels.

The concepts of Pareto optimality, stockout risk, and alternating modes of operation are fundamental to integrated systems management. Furthermore, these concepts are already supported by well-established computational tools, and can be efficiently adapted for mining and mineral processing operations. These tools and concepts may eventually become a valuable aid in the selection, sizing and operation of mining and mineral processing equipment.

3. Sample computations

This section analyzes concentrator tactics under geological uncertainty, considering a deposit that has two ore classes: Ore 1 and Ore 2. The concentrator had originally been designed with one operational mode, now called “Mode A”, which requires a 60-40 blend of ores 1 and 2. This mode has been applied successfully for several years. However, the geostatistical forecasts (based on core samples) now reveal that the deposit will give an average blend of 70-30 throughout the foreseeable future. Therefore, the perpetual application of Mode A would eventually cause an excess of Ore 1, and a shortage of Ore 2. This would constrain the mine production scheduling, and eventually lead to feed imbalances, decreased utilization of mining equipment, and costly decreases in downstream performance.

However, a new mode of operation “Mode B” has been developed through a series of bench-scale experiments and in-house plant trials. Mode B obtains a similar downstream performance as Mode A, except that it relies on smaller grinding media within the SAG mill, and is intended for an 85-15 blend. Table 2 contains data for the two modes; the intention is to consider changing modes during shutdowns. Assuming no stockout risk, Eqs. (2) and (3) determine that Mode A should

Table 2
Characterization of operational modes and deposit forecast.

	Mode A		Mode B		Deposit
	Regular	Contingency	Regular	Contingency	
Throughput (t/h)	250.0	162.5	225.0	104.2	–
Ore 1 in feed (%)	60	100	85	0	70
Ore 2 in feed (%)	40	0	15	100	30

be applied 1.37 times as often as Mode B, giving an average throughput of 239.44 t/day. Nonetheless, the implementation of Mode B is likely to require an upgrade in stockpiling equipment, which will benefit from bi-objective analysis, balancing the throughput and target stockpile level. If a maximal throughput can be maintained with small stockpiles rather than large stockpiles, this will imply (1) lower capital costs in terms of smaller-scale materials handling equipment, and (2) lower operating costs as smaller stockpiling heights imply less lifting of ore.

To maintain the total stockpile level, ore will be mined at 250 t/d under Mode A, and 225 t/d under Mode B. (It is assumed that the mining capacity exceeds the concentrator capacity, so that the concentrator functions as the bottleneck. Additionally, it is assumed that the mining method is relatively inflexible, e.g. block or sublevel caving (Adler and Thompson, 2011), hence the necessity of stockpiles to maintain feed consistency). Although the total stockpile level remains constant, the portion of Ore 1 is expected to increase under Mode A, and decrease under Mode B. Conversely, the portion of Ore 2 is expected to increase under Mode B, and decrease under Mode A. The decision to operate under Mode A or Mode B depends on the stockpile levels of the two ore classes, and is finalized during routine shutdowns that are performed every five weeks; the concentrator is then configured for the given operational mode for the entirety of the production campaign, which lasts 34 days (see Table 3).

According to a naïve deterministic analysis (similar to Eq. (1)), Mode B should be selected only when the level of Ore 2 is below 20.4 kt; this considers that stockpile 1 is expected to diminish by 0.6 kt/day under Mode A, and that the campaigns are 34 days, hence $0.6 \times 34 = 20.4$ kt. However, it does not account for geological uncertainty, and the resultant stockout risk. For instance, if the portion of Ore 2 coming from the orebody is unexpectedly low during Mode A, then the final days of the campaign may indeed suffer from a shortage of Ore 2, causing the plant to alternate between the regular and contingency mode; this gives a decrease in average throughput, as the contingency mode is substantially less productive than the regular mode (see Table 2). In this case, the contingency mode consumes only Ore 1, thereby allowing the Ore 2 stockpile to build up before reverting back to the regular mode. Similarly, if the portion of Ore 1 coming from the orebody is unexpectedly low during Mode B, then the end of the campaign may involve alternations between regular and contingency modes, corresponding again to a decrease in throughput. For simplicity, the contingency modes are assumed to have the same downstream performance as the regular modes, so that stockouts only cause a decrease in throughout and do not affect other indicators (recovery, reagent consumption, etc.). Table 3 includes a parameter that is called the Contingency Segment Duration, which has been set to 1 day, meaning that the contingency modes are applied for up to 1 day before reverting back to the regular modes. This parameter is illustrated in Fig. 4.

Table 3
Time segment parameters.

	Duration (days)
Production campaign	34
Shutdown	1
Contingency segment*	1

* Figs. 4–8 were produced using a contingency segment duration of 1 day, except for Fig. 4(b).

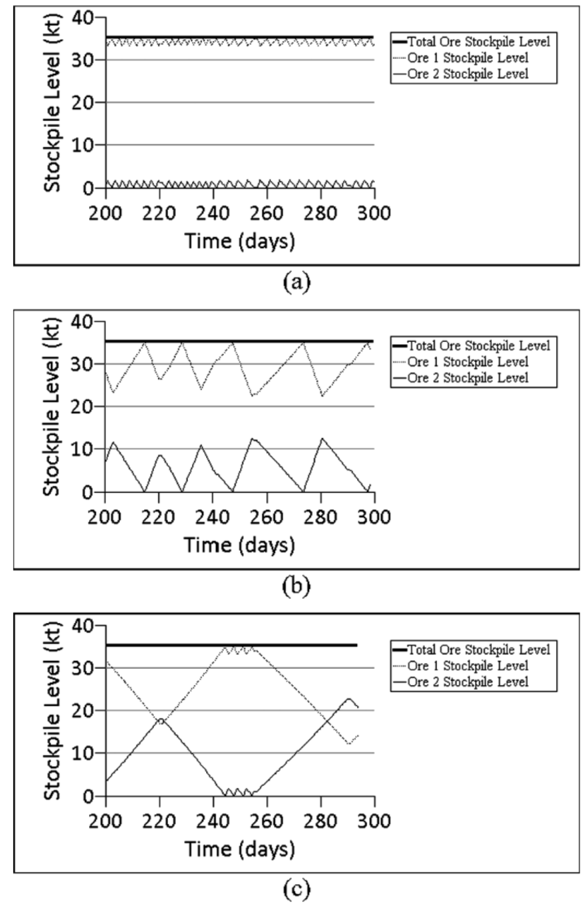
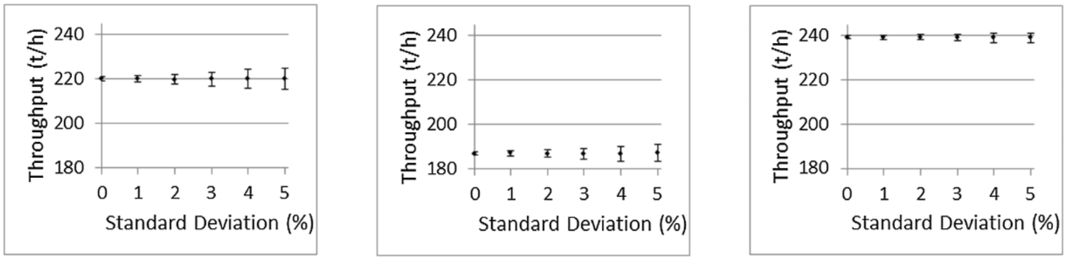


Fig. 4. Simulated operational dynamics, applying (a) only Mode A with contingency durations of one day, (b) only Mode A with contingency durations of one week, (c) a deterministic balance of Modes A and B with contingency durations of one day.

Fig. 4 and all subsequent computational results have been produced using a DES framework that was implemented in Rockwell Arena®, making extensive use of Visual Basic for Applications (VBA), and its dedicated implementation of the optimization engine OptQuest®; this engine relies on a combination of mathematical programming, metaheuristic and artificial intelligence techniques to search through possible system configurations to find the optimal configuration (Kleijnen and Wan, 2007). The framework was designed to optimize the operational response to geological uncertainty, by considering two control parameters: the critical transition level of Ore 2 and the total ore stockpile level. As described in the previous section, the first of these parameters is related to the reorder point R and the second of these parameters is related to the reorder quantity Q. The representation of geological uncertainty will be described below. The objective is to maximize throughput while minimizing the total stockpile level.

Fig. 4a and 4b consider only Mode A, which suffers from a chronic shortage of Ore 2. In both cases, the operations apply a contingency mode whenever the level of Ore 2 reaches zero. In Fig. 4a, the contingency mode is applied for one day, before reverting back to the regular mode which causes the Ore 2 level to fall back to zero, ultimately resulting in a fine saw-tooth trend. Fig. 4b exhibits a similar behaviour, except that the contingency mode is applied for durations of one week, forming a coarser saw-tooth shape. In either case, Eq. (5) predicts an average throughout of 220.34 t/h, as confirmed by Fig. 5a (described below). Fig. 4c demonstrates a balanced use of Modes A and B, which diminishes the occurrence of stockouts, thus increasing the average throughput.



(a) (b) (c)

Fig. 5. Simulated throughputs as a function of geostatistical uncertainty, applying (a) only Mode A, (b) only Mode B, and (c) a deterministic balance of Modes A and B assuming no stockout risk.

The framework incorporates geological uncertainty, as it reads data from an externally generated file. Moreover, geological uncertainty is the only source of uncertainty that is considered within the current implementation. The external file could contain results generated from a geostatistical software package, such as SGeMS (Remy et al., 2009). For demonstrative purposes, however, the current set of computations applies a simplified geostatistical model that determines the ore fraction from class 2, such that:

- Ore is mined from a single parcel at a time, before passing to the next parcel
- The amount of ore contained within each parcel is a random number that follows a uniform distribution, between 30 and 50 kt (this corresponds to roughly 1 week of production)
- Within a parcel, the ore fraction that is of class 2 is a random number that follows a Gaussian distribution, with mean 30% and standard deviation σ_{Geo}

The degree of geological uncertainty is controlled by varying the geostatistical standard deviation σ_{Geo}. The framework has been configured so that 600000 t of ore are processed within each replica, corresponding to over 1000 days of operation.

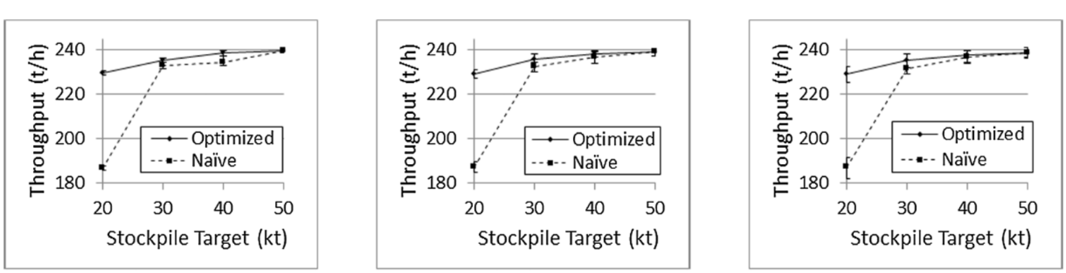
Fig. 5 contains simulated results for σ_{Geo} values of 0%, 1%, 2%, 3%, 4% and 5%, which agree with the deterministic analysis of Eqs. (2)–(5). Each of the data points was the result of 100 replicas, corresponding to over 100 000 simulated days per data point. The mean values shown in Fig. 5a are in agreement with Eq. (5) that predicts an average throughput of 220.34 t/h, irrespective of σ_{Geo}, when Mode A is used exclusively; however, increased geological uncertainty corresponds to increased throughput uncertainty, as the error bars become increasingly large as a function of σ_{Geo}. A similar observation can be made for Fig. 5b, in which Mode B is used exclusively, and the simulated means are compared to the analytical value of 186.45 t/h, which is obtained by adapting Eq. (5). Fig. 5c was generated using 20.4 kt as the critical Ore 2 level to decide between Modes A and B, and assuming sufficiently high total stockpile levels; the results are in agreement with Eq. (3), resulting in a mean throughput of 239.44 t/h.

As demonstrated in Fig. 6, it is risky to use 20.4 kt as the critical value when there is a possibility of stockout, i.e. when the total stockpile targets are well below 50 kt. For instance, the naïve application of this criterion when the stockpile target is 20.0 kt, causes the simulator to run exclusively with Mode B, giving a mean throughout of 186.45 t/h, as in Fig. 5b. However, even with these relatively low stockpile levels, the decision criteria can be optimized to give a result that is within 10 t/h of the de-risked value of 239.44 t/h, as in Fig. 5c. The optimized curves of Fig. 6 were obtained by applying the critical Ore 2 levels of Table 4 to 100 geological replicas as testing data. These critical Ore 2 levels were themselves obtained by fixing the total stockpile targets, and applying OptQuest with 10 geological replicas as training data. Similar trends are observed in Fig. 6a, 6b and 6c for σ_{Geo} values of 0%, 2.5% and 5%, respectively. As the total stockpile target level is increased towards 50 kt, stockout risk is abated, and the naïve and optimized curves converge. In practice, it is nonetheless preferable to adjust the operational management policies, instead of maintaining larger stockpiles that require larger equipment and greater operating costs.

Fig. 7 is based on the same data as Fig. 6b, which considers a total stockpile target of 30 kt, and a geostatistical standard deviation σ_{Geo} = 2.5%. The optimized naïve management policy uses a critical Ore 2 level of 20.4 kt, which is prescribed by the naïve deterministic analysis, whereas the optimized policy uses a value of 6.65 that is prescribed by OptQuest (see Table 4). The optimized policy is therefore more reluctant to apply Mode B. Mode A is indeed more productive under regular operation, 250 t/h versus 225 t/h (Table 2). Moreover, it is preferable to endure a stockout during Mode A rather than Mode B, considering a comparison of the contingency modes; therefore the portion of time in Mode B-Contingency (Fig. 7a) is largely replaced by Mode A-Contingency (Fig. 7b).

Fig. 8 is the Pareto frontier that describes the tradeoff between the target total stockpile level and the throughput that can be assured under a geological standard deviation of σ_{Geo} = 2.5%. This figure was generated by adapting a bi-objective procedure (Audet et al., 2012), using the values of Table 4 as initial solutions, and 10 geological replicas as training data (see Appendix). The simultaneously stochastically optimized values of critical Ore 2 level and total stockpile target are given in Table 5; these values were tested against 100 geological replicas, resulting in the dotted lines and error bars shown in Fig. 8.

Fig. 8 incorporates geological uncertainty and stockout risk, and can be used to evaluate different equipment upgrades that would support the



(a) (b) (c)

Fig. 6. Stochastic optimization of stockpile management policy with fixed total stockpile targets, assuming geostatistical standard deviations of (a) 0%, (b) 2.5%, (c) 5%

Table 4
Stochastically optimized critical Ore 2 levels for fixed total stockpile levels (kt).

Total stockpile level (kt)	20	30	40	50
σ_{Geo} = 0%	3.41	7.93	14.23	21.99
σ_{Geo} = 2.5%	3.12	6.65	15.48	20.46
σ_{Geo} = 5%	2.39	11.03	16.29	22.16

implementation of Mode B. For instance, equipment that is sized for stockpiles of 40 kt can ensure a throughput that is roughly 10 t/h higher than equipment that is sized for 20 kt; this comparison considers that the 40 kt and 20 kt options would each be operated using a correspondingly optimized critical Ore 2 level. The capital and operational costs that would be associated with the 40 kt option could perhaps be offset by the higher throughput. Otherwise, the 20 kt option would be preferred.

4. Conclusions and future work

Ore deposits are usually heterogeneous, forcing the concentrator feed to evolve over time. The nature of this evolution should be considered early within the development of a mine, but may need to be revisited at critical times during the life of the operation. Sections of the orebody are indeed mined, stockpiled and processed with due consideration of the available resources, the capabilities of the plant, the sequential mining constraints and the specifications of the outgoing concentrates. It is common practice to create and manage stockpiles with different ore classes; however, there are standard optimization tools that have not yet been fully adapted for ore stockpile management.

In this paper, a simulation of the stockpile mine production sequence is considered in relation to the classic RQ inventory problem, in which the stockpiles are considered as the inventory level, and the mode-changing events are analogous to the reorder events. Although the problem has been simplified to consider only two ore types, the framework could be extended to a larger number of ore types that could come from a range of geological domains in a mine. The RQ problem could then be extended to determine when to start mining particular types of ore in a mine plan, so as to minimize the impact on the plant throughput commensurate with the target cash flow. Posed in this way, the mine-to-mill and mine planning exercise becomes the same problem as stocking a warehouse or a shop, and is amenable to the solutions derived in these disparate applications.

Although the quantity of valuable mineral in an ore is of interest at the separation stage of a plant, the size reduction of mined ore is fundamental to obtaining the required liberation and throughput. The primary grinding mills and their infrastructure can be major bottlenecks within this process. A number of variables are available to affect mode changes, which include the usual mill speed and water addition rate. Although the mill throughput is

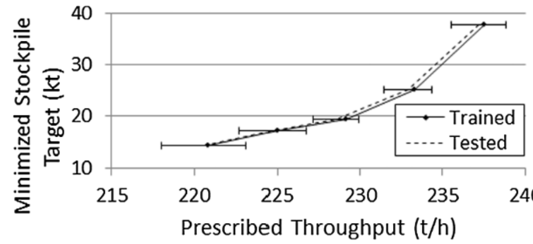


Fig. 8. Pareto frontier characterizing tradeoff between throughput and stockpile target level, given a geostatistical standard deviation of 2.5%

Table 5
Stochastically optimized operational management parameters for σ_{Geo} = 2.5%

Prescribed throughput (t/h)	220.83	225.00	229.17	233.33	237.50
Minimized total stockpile target (kt)	14.32	17.30	19.48	25.12	37.75
Corresponding critical Ore 2 level (kt)	0.15	0.69	1.11	2.09	6.74

adjustable, it is common to leave the adjustment of this variable as a last resort. Ball addition rate is also available for adjustment, but there is asymmetry in the adjustment of the ball load inside the mill, since it takes some time for the ball level to decrease compared to the possibly instantaneous increase in ball level. A semi-permanent adjustment of mill geometries is an even longer term consideration (Chimwani et al., 2013), left for dedicated campaigns of different ore types, as distinct from the relatively fast mode changes envisioned in this paper. Other fast adjustments to the grinding circuit could modify crusher performance and, more recently, HPGR performance. Ultimately, the adjustments of the grinding parameters must be in response to the incoming mineralogy, in consideration of the performance of downstream separation processes (Suazo et al., 2010; Chimwani et al., 2013). Specifically, a better utilization of grinding energy is possible through system-wide coordination (Carrasco et al., 2017; Lotter et al., 2018), and the definition of alternating modes of operation that respond to changing stockpile level.

The current paper is focused on geological uncertainty, although real mines may suffer from operational problems such as equipment failures, or unforeseen delays in receiving consumables (explosives, blasting agents, machine parts, etc.) that would hinder mine production. These aspects could be directly included in the current model, as temporarily changes in the stockpile rates; the stockpiles would eventually be replenished after the operational problems are resolved, notwithstanding the short-term risk of stockout. Eqs. (2)–(5) and their multi-stockpile extensions remain valid as deterministic benchmarks. Moreover, the incorporation of models for the unit processes of a

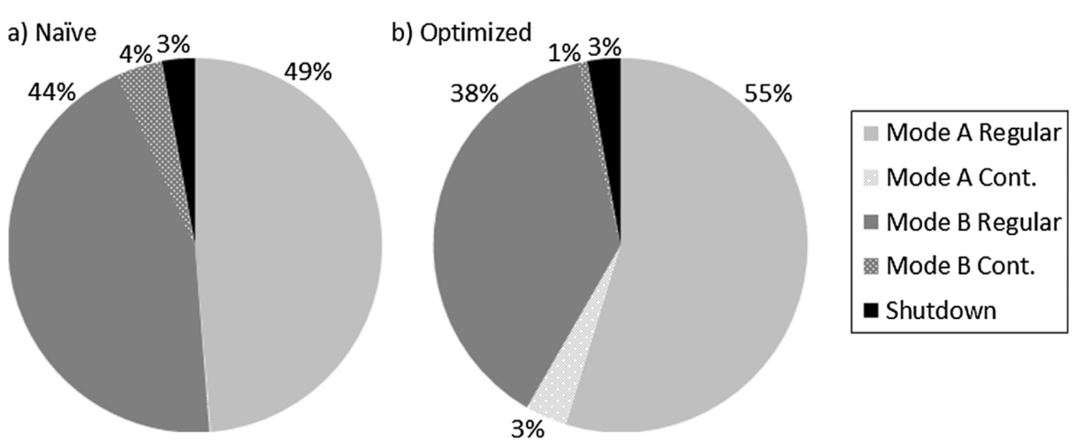


Fig. 7. Time-averaged distribution of operational modes for a total stockpile target of 30 kt, assuming a geostatistical standard deviation of 2.5%, for (a) deterministic balance of Modes A and B, and (b) stochastically optimized operational management policy.

Table 6
Multistage minimization of target stockpile level while maintaining prescribed throughput.

	Initial solution		Optimization		Final solution	
Prescribed throughput (t/h)	Critical Ore 2 level (kt)	Target stockpile level (kt)	Number of stages	Final Interval	Critical Ore 2 level (kt)	Target stockpile level (kt)
237.50	15.48	40.00	2	[35.00, 37.50]	10.80	36.91
233.33	10.80	36.91	5	[25.00, 27.50]	8.16	25.03
229.20	8.16	25.03	4	[17.50, 20.00]	2.60	19.53
225.00	2.60	19.53	2	[15.00, 17.50]	2.01	16.40
220.83	2.01	16.40	2	[12.50, 15.00]	1.56	13.32

mineral processing plant, together with the sequential and mode change simulation presented in this paper, present an opportunity to effectively model the mine-to-mill sequence from a systemic perspective, with the benefit of obtaining mine-local and global optimizations for cash flows and asset utilization. Additional benefits would accrue from using these simulations in real-time and perhaps within a supervisory control framework. Extending the approach of Suazi et al. (2010) and Lotter et al. (2018), these types of realistic problems will be addressed, in conjunction with geological uncertainty, through the development of industrial case studies.

Appendix A

Construction of a pareto frontier using a black box optimization engine

Black box optimization engines such as OptQuest provide an effective means to develop quantitative frameworks to support engineering decisions, including system design and the adjustment of operational policies. Within the early phases of an engineering project, a black box engine can be included within a prototype of a framework and, in later phases, the engine may be replaced by customized algorithms. The drawback of black box optimization is that the underlying mathematical and computational principles are largely confidential; it is then difficult to assess what exactly are the limitations of the engine, especially for the simultaneous optimization of severable variables. The responsible development of these frameworks requires firstly the replication of simplified results that can be verified. For instance, the average values observed in Fig. 5 are in agreement with the results of Eqs. (2)–(5); additionally, the size of the confidence intervals of Fig. 5 increases as a function of the geostatistical standard deviation, as expected. Only after having verified that the framework can function correctly under simplified configurations, is it appropriate to deviate toward more demanding and/or realistic configurations.

During preliminary experimentation with OptQuest to optimize a single variable, the engine was found to be most effective if the user specifies an interval of exploration for that variable. This is consistent with an evolutionary optimization process (EVOP). Therefore, the construction of Fig. 8 employed the following strategy to locate minimal target stockpile values that maintain a prescribed throughput:

- The optimization consists of one or more stages, which are each confined to an exploration interval that has width 2.50 kt. Moreover, each of these stages is to measure performance with respect to ten operational scenarios (training data).
- As a result of an optimization stage, the engine may suggest configurations that are clustered at the lower boundary of the exploration interval. In the next stage, the exploration interval is therefore shifted down by 2.5 kt; the suggested results of the previous stage are now used as initial solutions for the next stage.
- If, during a given stage, the minimal target stockpile value is within the interval rather than at the lower bound, then this value is accepted as the minimum value, and there are no additional stages to be performed.

For instance, the first value of Table 6 considers a target stockpile value of 40 kt, which is capable of maintaining a throughput of 237.5 t/h, using a critical ore value of 15.48 kt (See Table 4). OptQuest is then used to search for configurations with target stockpile values as low as 37.50 kt, but which still maintain a throughput of 237.5 t/h. As a result of this search, OptQuest makes suggestions clustered at 37.50 kt, which are then used as initial solutions in the second stage, to seek values as low as 35.00 kt. This second stage only reaches as low as 36.91 kt, and so there is no need for a third stage. Nonetheless, this configuration is used to initiate the second row of Table 6. Subsequently, the second row requires five stages, and finds a minimal stockpile target of 25.03 kt, which is then used to initiate the third row, and so on.

Within Table 6, the critical Ore 2 levels seem to follow a downward trend. However, the aforementioned minimization did not include any explicit restrictions on the critical Ore 2 level, which seems to suggest an untapped degree of freedom (a.k.a. degeneracy). OptQuest was therefore used to refine the results of Table 5; this involved minimizing the critical Ore 2 level while maintaining approximately the same throughput and target stockpile levels, again using ten randomly generated operational scenarios; the refined results are given in Table 4. For instance, the unrefined critical Ore 2 level for 220.83 t/h is 1.56 kt (Table 5), whereas the refined value 0.15 kt (Table 4) is nearly zero. Indeed, the refined value is in agreement with Fig. 5a and Eq. (5), that such a throughput can be maintained without Mode B.

The current paper presents an adaptation of the two-mode inventory model that is computationally simple, from the optimization point of view. The maximal throughput is a nondecreasing function of the target stockpile level and, likewise, the minimal total stockpile level is a nondecreasing function of prescribed throughput. OptQuest is apparently capable of detecting these relationships automatically, and applying binary searches that subdivide the intervals of Table 6. It is unclear what exactly OptQuest does to simultaneously vary two or more parameters, however Audet et al. (2012) describe techniques that are based on Mesh Adaptive Direct Search that are appropriate for simulation-based optimization.

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